

PNEUMONIA DETECTION FROM CHEST X-RAYS USING VGG16

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ABSTRACT

This study fine-tunes VGG16 on the Kaggle Chest X-Ray dataset for binary pneumonia classification, achieving 95.5% accuracy and 0.97 AUC-ROC, demonstrating potential for supporting radiologist workflows.

INTRODUCTION

Pneumonia is a leading cause of hospitalization in the Philippines, yet radiologist availability in provincial hospitals is limited. Deep learning models can assist in chest X-ray interpretation, flagging pneumonia cases for priority review. This study fine-tunes VGG16 on a publicly available chest X-ray dataset and evaluates clinical viability.

METHODOLOGY

VGG16 pretrained on ImageNet was fine-tuned on the Kaggle Chest X-Ray Images (Pneumonia) dataset (5,856 images: 1,583 Normal, 4,273 Pneumonia). Class imbalance was addressed by class-weighted loss (weight ratio 2.7:1). Training: 25 epochs, SGD optimizer ($\text{lr}=0.001$, momentum=0.9), batch size 32, image size 224×224 . Augmentation: random rotation, width/height shift, zoom, horizontal flip. Evaluation: accuracy, AUC-ROC, sensitivity, specificity.

RESULTS

VGG16 achieved 95.5% accuracy, AUC-ROC 0.97, sensitivity 97.4%, and specificity 91.3%. False negatives (missed pneumonia) numbered 11 in the test set (4 bacterial, 7 viral). Grad-CAM visualizations confirmed the model attended to clinically relevant consolidation regions. Inference time: 45ms per image.

DISCUSSION

VGG16 transfer learning effectively detects pneumonia with high sensitivity, prioritizing recall to minimize missed cases. The 91.3% specificity is acceptable for a screening tool where further human review follows flagging. Grad-CAM validation builds clinician trust. Future work should validate on Filipino patient chest X-rays and evaluate under PACS integration in a provincial hospital setting.